

CS543 Web Services — Assignment 3

Integrate an External SOAP Partner — SCM (Smart Contact Manager)

Team ID: 12

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1. Context

Partner: **QuickPing SMS Gateway** (an external SMS provider). Operation integrated: **SendSms** — submit one text message to a single contact phone number (E.164). SCM stores users' contacts, including phone numbers, and can provide a planned SMS action for a selected contact. The browser-facing side of SCM remains REST/JSON, while the external SMS edge is SOAP because the partner publishes a strongly typed WSDL contract, supports message-level credentials, and exposes typed faults and retry semantics.

2–3. Artefacts

The submission includes partner.wsdl and the three SOAP envelopes: soap-request.xml, soap-response.xml, and soap-fault.xml. The envelopes are designed to match the WSDL element-for-element.

4. HTTP Binding

POST /soap/v1/messaging HTTP/1.1

Host: api.quickping-sms.com

Content-Type: text/xml; charset=utf-8

SOAPAction: "http://schemas.quickping-sms.com/messaging/v1/SendSms"

Endpoint: https://api.quickping-sms.com/soap/v1/messaging

Success: HTTP 200 OK **Fault:** HTTP 500 Internal Server Error

5. Discovery

SCM obtains the WSDL from the partner onboarding URL: https://api.quickping-sms.com/soap/v1/messaging?wsdl. The WSDL is obtained at build time and the runtime endpoint is kept in configuration. A lightweight catalogue record is used instead of running a live UDDI server.

Business: QuickPing Communications Pvt. Ltd.

Service: QuickPingMessagingService

Binding: MessagingSoapBinding (SOAP 1.1 / HTTP, document-literal)

Endpoint: https://api.quickping-sms.com/soap/v1/messaging

WSDL: https://api.quickping-sms.com/soap/v1/messaging?wsdl

Operation: SendSms

Authentication: AuthHeader (AccountId + ApiKey)

6. Fault Mapping

The planned SCM endpoint **POST /api/contacts/{id}/sms** would be the only SCM-facing component that knows the partner protocol. A translator maps partner fault codes into SCM's own public errors so partner vocabulary and credentials never leak to SCM users.

Partner fault	SCM HTTP / error	User-facing message	Internal action
INVALID_MSISDN	422 CONTACT_PHONE_INVALID	This contact's phone number is not valid. Please edit the contact and try again.	None — user can fix the phone number.
RATE_LIMITED	429 TOO_MANY_MESSAGES	You are sending messages too quickly. Please wait a moment.	Retry later; Retryable=true.
GATEWAY_UNAVAILABLE	503 SMS_TEMPORARILY_UNAVAILABLE	We could not send your message right now. Please try again shortly.	Retry with the same ClientReference.

Partner fault	SCM HTTP / error	User-facing message	Internal action
INSUFFICIENT_CREDENTIALS	503 SMS_TEMPORARILY_UNAVAILABLE	We could not send your message right now. Please try again shortly.	Alert SCM administrator; not the user's fault.
AUTH_FAILED	503 SMS_TEMPORARILY_UNAVAILABLE	We could not send your message right now. Please try again shortly.	Alert SCM administrator; rotate the credential.

Worked example: soap-fault.xml contains INVALID_MSISDN for the demonstration number. SCM maps it to HTTP 422 with CONTACT_PHONE_INVALID. Raw partner faults and demonstration credentials remain server-side only.

Submission

Required artefacts: integration.pdf, partner.wsdl, soap-request.xml, soap-response.xml and soap-fault.xml. Team ID 12 and both team members are included above. Assignment 3 is design-time only; no running SOAP server or live gateway is required.